PRECALCULUS · UNIT 6: MATRICES

Spy Agency Codes

Use matrices to manage inventory, encode secret messages, and solve systems.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Welcome to the Cipher Division. Agents carry gadgets from three warehouses, field reports must be encoded so enemies cannot read them, and supply problems arrive as systems of equations.

You will use matrices to track gadgets, encode and decode messages, and solve supply problems.



Key Matrix

K = [[3, 1], [5, 2]]

Alphabet

A = 1, B = 2, …, Z = 26

space = 27

Learning Goals

- I can add matrices and multiply by a scalar.
- I can multiply a matrix by a column matrix.
- I can encode a message with a key matrix.

Project Timeline

When	What you will do
Day 1	Inventory
Day 2	Costs
Day 3	Secret Codes
Final	Agent Mission File + reflection

Part 1: Inventory

Day 1 · Write the inventory as a 3 × 3 matrix I (rows Boston, London, Tokyo; columns pens, watches, drones).

Gadget Inventory (rows: warehouses, columns: gadgets)

Boston	London	Tokyo
Pens 12 · Watches 5	Pens 8 · Watches 9	Pens 15 · Watches 6
Drones 3	Drones 4	Drones 7

Costs	New Shipment	Key Inverse
Pen $40 · Watch $250	Each warehouse gets	$K^{-1} = [[2, -1], [-5, 3]]$
Drone $900	+2 pens, +1 watch, +1 drone	

1. Write matrix I and give its dimensions.

2. Add the New Shipment matrix S = [[2, 1, 1], [2, 1, 1], [2, 1, 1]] to I.

3. The agency doubles the original inventory. Find 2I.

4. What does the entry in row 3, column 3 of I mean?

Part 2: Costs

Day 2 · Cost column matrix C = [40, 250, 900] (pens, watches, drones).

5. Find the value of the Boston gadgets: 12(40) + 5(250) + 3(900).

Starter: *12 × 40 + 5 × ___ + 3 × ___*

6. Find I · C (the value of each warehouse).

7. Which warehouse has the most valuable inventory?

8. Why can we multiply a 3 × 3 matrix by a 3 × 1 matrix?

Part 3: Secret Codes

Day 3 · Key K = [[3, 1], [5, 2]]. Letters: A = 1, …, Z = 26.

9. Write the word HI as a column matrix.

10. Encode HI by computing K · [8, 9].

11. Decode [33, 58] using K^{-1} = [[2, −1], [−5, 3]]. Do you get HI back?

Final Product: Agent Mission File

Your final product must include:

- Inventory matrices and operations
- Warehouse values
- Encoded and decoded message
- Explanation of dimensions
- Mission file

Reflection

R1. Why do the dimensions have to match to add matrices?

R2. Why does the inverse matrix decode the message?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Matrix operations, inverses, and determinants are correct and used for inventory, codes, and systems. Goes beyond the minimum with a creative, realistic choice.	Matrix operations, inverses, and determinants are correct and used for inventory, codes, and systems.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Mission file includes matrices, coded messages, solved systems, and explanations. Polished, organized, and visually engaging.	Mission file includes matrices, coded messages, solved systems, and explanations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.